Home / Grade 10 / Algebra / Quarter 1

Quadratic inequalities and simultaneous equations

Cambridge insert: solving with a sketch rather than a rule, and the substitution that handles a curve meeting a line.

LESSON 6–7 | 2 × 40 MIN | ALGEBRA 10, REVISION | P1 · 1.2, 1.7

LESSON CLOCK

Sketch three parabolas and shade where y > 0	6 min
Quadratic inequalities	14 min
Simultaneous equations	12 min
When does the line miss?	4 min
Homework	4 min

LEARNING OBJECTIVES

- Solve a quadratic inequality by sketching the parabola.
- Write the answer with the right connective — “and” or “or”.
- Solve one linear and one quadratic equation together.
- Use the discriminant to decide whether a line meets a curve.

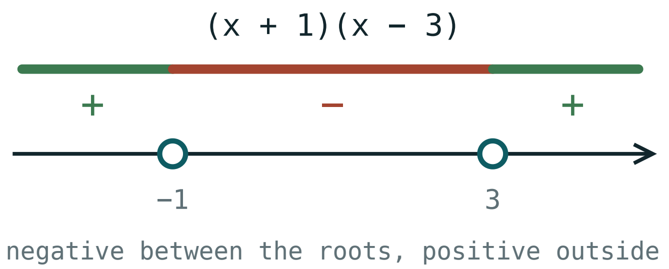
TEXTBOOK

National	Повторение, pp. 9–13
Cambridge	Pure Mathematics 1, pp. 6, 16–17
Workbook	P1 Exercise 1C, 1G

01 Explanation

Inequalities: sketch, do not memorise

To solve $x^2 - 2x - 3 > 0$: find where the expression is **zero**, sketch the parabola, then read off where it is above the axis.



The critical values split the line into three. A parabola opening upwards is negative only between its roots.

THE METHOD, IN THREE STEPS

1. Make one side zero and factorise: $(x + 1)(x - 3) > 0$.
2. Mark the critical values -1 and 3 on a sketch of the parabola.

3. Read the answer off: above the axis **outside** the roots, so $x < -1$ or $x > 3$.

THE CONNECTIVE CARRIES A MARK

Between the roots the answer is **one** interval and takes “and”: $-1 < x < 3$. Outside the roots it is **two** intervals and takes “or”: $x < -1$ or $x > 3$. Writing $3 < x < -1$ says nothing exists.

One linear and one quadratic

Always substitute from the **linear** equation into the quadratic — never the other way round. Solve $y = x + 1$ with $x^2 + y^2 = 25$:

$$x^2 + (x + 1)^2 = 25 \Longrightarrow 2x^2 + 2x - 24 = 0 \Longrightarrow x^2 + x - 12 = 0$$

So $x = 3$ or $x = -4$, and the line $y = x + 1$ gives $y = 4$ and $y = -3$. Two intersection points: $(3, 4)$ and $(-4, -3)$.

PAIR THE ANSWERS UP

Each x belongs with **its own** y. Listing $x = 3, -4$ and $y = 4, -3$ separately loses the pairing, and the mark.

Does the line meet the curve at all?

After the substitution you have a quadratic. Its discriminant answers the geometry:

D	THE LINE AND THE CURVE
$D > 0$	meet at two points
$D = 0$	touch — the line is a **tangent**
$D < 0$	never meet

So “find m so that $y = mx + 3$ is a tangent to $y = x^2$” is not a geometry question at all. Substitute, set $D = 0$, solve.

02 Worked examples

EXAMPLE 1 **Solve $x^2 - x - 6 \le 0$.**

1. $(x - 3)(x + 2) \le 0$

   Factorise.

2. *critical values* -2 *and* 3

3. The parabola opens upwards, so it is at or below the axis **between** the roots.

4. $-2 \le x \le 3$

   Closed circles: $\le$ includes the ends.

ANSWER

$-2 \le x \le 3$

EXAMPLE 2 **Solve $y = 2x - 1$ together with $y = x^2 - 4$.**

1. $2x - 1 = x^2 - 4$

   Both equal y.

2. $x^2 - 2x - 3 = 0$

3. $(x - 3)(x + 1) = 0$

4. $x = 3 \implies y = 5;\ x = -1 \implies y = -3$

   Pair them.

ANSWER

$(3, 5)$ and $(-1, -3)$

EXAMPLE 3 **Find c so that $y = 3x + c$ is a tangent to $y = x^2 + 5$.**

1. $3x + c = x^2 + 5$
2. $x^2 - 3x + 5 - c = 0$
3. $D = 9 - 4(5 - c) = 0$

   Tangent means one repeated root.
4. $9 - 20 + 4c = 0$
5. $c = \dfrac{11}{4}$

ANSWER

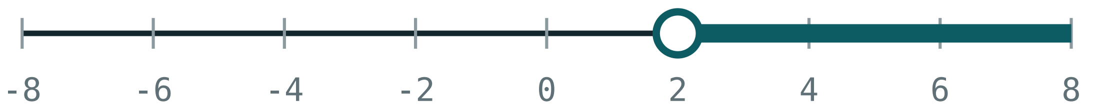
$c = \dfrac{11}{4}$

03 Interactive model

Move the coefficients; watch which region of the number line makes the parabola positive.

Where is the expression positive?

$x > 2$



inequality `x > 2`	boundary `2 (open)`	x = 0 works? **no**	x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Quadratic inequality	Kvadrat tengsizlik	Квадратное неравенство
Critical value	Kritik qiymat	Критическое значение
Sign chart	Ishoralar jadvali	Схема знаков
Sketch	Eskiz	Эскиз
Simultaneous equations	Tenglamalar sistemasi	Система уравнений
Substitution method	O‘rniga qo‘yish usuli	Метод подстановки
Point of intersection	Kesishish nuqtasi	Точка пересечения
Tangent	Urinma	Касательная
Interval	Oraliq	Промежуток

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$(x - 1)(x - 4) < 0$ gives:

- $x < 1$ or $x > 4$
- $1 < x < 4$
- $x > 4$
- $x < 1$

$x^2 > 9$ gives:

- $-3 < x < 3$
- $x > 3$
- $x < -3$ or $x > 3$
- $x > 9$

Substituting a line into a curve gives $D < 0$. They:

- touch
- meet twice
- never meet
- are the same

Solving $y = x$ with $y = x^2$ gives:

- $(0,0)$ only
- $(1,1)$ only
- $(0,0)$ and $(1,1)$
- no points

Which is impossible?

- $2 < x < 5$
- $x < 2$ or $x > 5$
- $5 < x < 2$
- $x \geq 5$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Solve $(x - 2)(x - 5) < 0$

   ANSWER $2 < x < 5$

2. Solve $(x + 1)(x - 3) > 0$

   ANSWER $x < -1$ or $x > 3$

3. Solve $x^2 < 16$

   ANSWER $-4 < x < 4$

4. Solve $x^2 \geq 25$

   ANSWER $x \leq -5$ or $x \geq 5$

5. Solve $x^2 - 4x \leq 0$

   ANSWER $0 \leq x \leq 4$

6. Solve $y = x + 2$ and $y = 2x$

   ANSWER $(2, 4)$

7. Solve $y = x^2$ and $y = 4$

   ANSWER $(\pm 2, 4)$

Medium · the standard question

7 PROBLEMS

1. *Solve* $x^2 - 5x + 6 > 0$

   ANSWER $x < 2$ or $x > 3$

2. *Solve* $2x^2 + x - 6 \leq 0$

   ANSWER $-2 \leq x \leq 1.5$

3. *Solve* $6 - x - x^2 > 0$

   ANSWER $-3 < x < 2$

4. *Solve* $y = x - 1$ *and* $y = x^2 - 3$

   ANSWER $(2, 1)$ and $(-1, -2)$

5. *Solve* $y = 2x$ *and* $x^2 + y^2 = 20$

   ANSWER $(2, 4)$ and $(-2, -4)$

6. *Find* c *so* $y = 4x + c$ *is a tangent to* $y = x^2$.

   ANSWER $c = -4$

7. *Solve* $x^2 + 3x \geq 10$

   ANSWER $x \leq -5$ or $x \geq 2$

Hard · stretch and reason

7 PROBLEMS

1. Solve $x^2 - 2x + 5 > 0$

ANSWER every x — $D < 0$ and the parabola opens upwards

2. Solve $x^2 + 4x + 4 \le 0$

ANSWER $x = -2$ only

3. Solve $\dfrac{x - 1}{x + 2} > 0$

ANSWER $x < -2$ or $x > 1$

4. Find m so $y = mx - 2$ is a tangent to $y = x^2 + 2$.

ANSWER $x^2 - mx + 4 = 0$, $m^2 = 16$, $m = \pm 4$

5. Solve $y = 3 - x$ and $x^2 + y^2 = 5$

ANSWER $(1, 2)$ and $(2, 1)$

6. For what k does $y = x + k$ never meet $y = x^2$?

ANSWER $x^2 - x - k = 0$, $1 + 4k < 0$, so $k < -\dfrac{1}{4}$

7. Solve $x^4 - 5x^2 + 4 < 0$

ANSWER $t = x^2$: $1 < t < 4$, so $1 < |x| < 2$ — that is $-2 < x < -1$ or $1 < x < 2$

07 Homework

Homework — 6 problems

P1 Exercise 1C and 1G. Sketch the parabola for every inequality.

1. Solve $(x - 1)(x + 4) < 0$
2. Solve $x^2 - 7x + 10 \ge 0$
3. Solve $3 + 2x - x^2 > 0$
4. Solve $y = 2x + 1$ and $y = x^2 - 2$

5. *Solve* $y = x - 2$ *and* $x^2 + y^2 = 10$

6. *Find* c *so that* $y = 2x + c$ *is a tangent to* $y = x^2 + 1$.